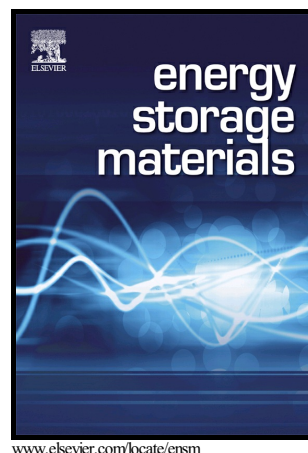


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Plastic crystal polymer electrolytes containing boron based anion acceptors for room temperature all-solid-state sodium-ion batteries

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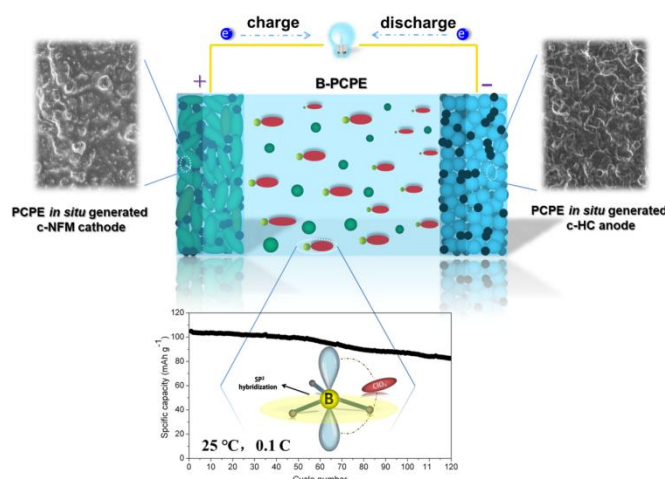
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Abstract

It is always a compelling challenge to develop solid electrolyte for ambient-temperature all-solid-state rechargeable batteries, which occupies superior ionic conductivity, high ion transference number, considerable mechanical property, and favorable interfacial contact. Here, a nonwoven supported plastic crystal polymer electrolyte containing anion-trapping boron moieties (B-PCPE) for all-solid-state sodium-ion batteries (SIBs) is first reported to improve overall performances. The B-BCPE was prepared by *in situ* growth of plastic crystal electrolyte and boron-containing cross-linker inside a nonwoven support via UV-curing technique. With an anion acceptor contained three-dimensional network structure, the B-PCPE simultaneously exhibited a remarkable room temperature ionic conductivity (0.36 mS cm^{-1}), high sodium-ion transference number (~ 0.62), and a superior tensile strength (28.2 MPa). Remarkably, plastic crystal polymer electrolyte impregnated composite $\text{NaNi}_{1/3}\text{Fe}_{1/3}\text{Mn}_{1/3}\text{O}_2$ (c-NFM) cathode and hard carbon (c-HC) anode were designed by *in situ* growth technique. By this technique, ion transport in electrodes as well as that between electrode-electrolyte interfaces could be strengthened. Therefore, the cycle and rate performance of all-solid-state SIB assembled by this technique are greatly enhanced. The all-solid-state SIB assembled by this technique delivers a high first discharge capacity of 104.8 mAh g^{-1} with capacity retention of about 80.1% after 120 cycles at 0.1 C. It's believed that B-PCPE is a promising candidate for practical application and provides a new perspective to design high-performance solid

electrolyte for ambient-temperature all-solid-state batteries with exceptional rate capability and cycle stability.

Graphical abstract



Keywords: Plastic crystal polymer electrolyte; boron based anion acceptor; all-solid-state; favorable interfacial contact; room-temperature sodium-ion battery

1. Introduction

Sodium-ion batteries (SIBs) are being actively investigated as promising alternatives to lithium-ion batteries (LIBs) for the next-generation large-scale energy storage devices due to low cost, high natural abundance and uniform worldwide distribution of Na resources [1-3]. However, limited energy density and increasing safety issues due to the use of conventional flammable organic liquid electrolytes in large-scale batteries are of great concern [4]. Replacing organic liquid electrolytes with solid electrolytes in all-solid-state SIBs can fully address such safety issues and in parallel accompany with high energy density. As the key component in all-solid-state SIBs, the solid electrolytes have been intensively studied and mainly include inorganic Na⁺ conductors (INCs) and solid polymer electrolytes (SPEs). Currently, the greatest

bottlenecks in developing all-solid-state SIBs are the inherent relatively poor ambient-temperature ionic conductivity, especially for the SPEs, and the sluggish interface charge-transfer caused by the large interfacial resistance between solid electrolyte and electrodes, especially for the INCs [5, 6]. Although many efforts have been made, the all-solid-state batteries still can't meet the requirement of practical application, and some can only work at high temperature but fail to conduct Na^+ at ambient temperature [7-9]. Therefore, it's urgent to develop all-solid-state sodium ion batteries with perfect electrochemical performances at room temperature.

Plastic crystal electrolytes (PCEs) are a new type of fast ion conducting solid electrolytes, which is composed of metal salt and polar plastic crystal with excellent solvating capability. Succinonitrile (SCN) is a typical non-ionic plastic crystal and shows a plastic-crystalline phase in the temperature range from $-38\text{ }^{\circ}\text{C}$ to its melting point (about $60\text{ }^{\circ}\text{C}$), which allows trans-gauche isomerism involving rotation of molecules about central C-C bonds of SCN. Moreover, the terminal nitrile groups attract metal ions not so strongly, which will facilitate charge separation of different metal salts [10-13]. Thus the SCN based plastic crystal electrolytes exhibit a remarkable ionic conductivity over 10^{-3} S cm^{-1} at room temperature. SCN has already been widely used as electrolyte host, solid plasticizer or additive in lithium-ion batteries [14-18], fuel cells [19] and so on, due to the additional non-flammability and superior thermal stability (stable up to approximately $150\text{ }^{\circ}\text{C}$). At the same time, it is reported that the size of the mobile metal cation is not a limiting factor, and even a higher ionic conductivity of Na^+ than that of Li^+ in a SCN based plastic crystal electrolyte has already been demonstrated [14, 20]. However, up to now, there are few papers reported on the SCN based PCEs for all-solid-state SIBs with acceptable performances. One of the most important reasons is that the mechanical properties of SCN based PCEs are very poor, which greatly hinder its application in rechargeable batteries. Generally, the SCN based PCEs are excessively soft and easily deform under low stress.

In the past decades, many efforts have been done to overcome this shortcoming of SCN based PCEs. Among these, the design of plastic crystal polymer electrolytes

(PCPEs) by combining the PCEs with a polymer matrix is widely recognized as an effective and facile approach in which the polymer matrix is chosen as a mechanical reinforcing framework [15, 21-23]. In PCCEs, there is a balance between their mechanical properties and electrochemical performance. Increase of SN concentration in PCPEs is favorable to deliver high ionic conductivity but may also deteriorate the mechanical strength of the PCPEs. Although the PCPEs obtained from above approaches showed improved mechanical properties compared to that of original PCEs, most PCPEs are still accompanied by insufficient mechanical flexibility. In order to obtain PCPEs with high ionic conductivity as well as mechanical strength, frameworks composed of nonwovens are adopted to make up the poor mechanical flexibility. In addition, as with conventional organic liquid electrolytes, SCN based PCEs also showed a low ion transference number due to the reverse mobility of anion, and the cation, Na^+ , carries a more stable solvation shell than the anions, the cation transference numbers (t_{Na^+}) are always inferior (often ca. 0.2-0.4 for the cation) [24]. This will cause concentration gradients of the salt and cell polarization during the charge/discharge process, resulting in deterioration of long term cycle performance and power density [25-28]. Introducing anion receptors in electrolytes will effectively improve ion transference number, because they can selectively complex with anions in electrolytes, thereby reduce the mobility of anions [29, 30]. Polymers with boron moieties are promising anion receptors in electrolytes thanks to the Lewis acidity of the elements of B, which effectively capture the basic anions from electrolyte salts and can facilitate dissociation of the salts and simultaneously increase the ion transference number of the electrolytes [31-34]. To date, as anion acceptor, solid electrolyte containing boron moieties have been investigated in lithium-ion batteries [28, 31, 34, 35], symmetrical capacitors [36], etc. However, boron-based anion acceptors have not been reported for Na battery application.

In the present study, we have successfully synthesized a new type of anion acceptor-contained plastic crystal polymer electrolyte for ambient temperature sodium-ion batteries by *in-situ* UV-curing of PCE and boron-containing cross-linker (B-HEMA) inside a polypropylene-cellulose composite nonwoven (PCN) skeleton.

Herein, the PCN was a mechanical reinforcing framework, and the UV-crosslinked B-HEMA polymer in B-PCPE was design to simultaneously improve mechanical and electrochemical performances. Based on this unique structural/composition integrity, the B-PCPE showed remarkable improvement in room-temperature ionic conductivity, ion transference number, as well as mechanical properties. Furthermore, c-NFM cathode and c-HC anode that all contain similar chemical component as B-PCPE were designed and prepared by *in situ* growth technique to form favorable electrode-electrolyte interfaces, which can benefit the electrochemical performance of all-solid-state SIB.

2. Experimental

2.1. Materials

Trimethyl borate (TMB, $\geq 99.8\%$), 2-hydroxyethyl methacrylate (HEMA), 2-hydroxy-2-methyl-1-phenyl-1-propanone (HMPP), and poly (vinylidene fluoride) (PVDF, $M_w = 534000$ g/mol) were purchased from Aldrich and used as received. Succinonitrile (SCN, 99.0%) was obtained from J&K Scientific Co. Ltd. Sodium perchlorate (NaClO_4 , 99.99%), trimethylolpropane ethoxylate triacrylate (ETPTA, $M_n=428$ g/mol), anhydrous acetonitrile (AN, 99.8%), and N-methyl-2-pyrrolidone (NMP, 99.5%) was purchased from Aladdin Chemical Reagent Co. Ltd. NaClO_4 were dried under vacuum at $130\text{ }^\circ\text{C}$ for 48 h prior to use and stored in an argon filled glove box. A polypropylene-cellulose composite nonwoven (PCN) was a gift from Saibo Materials.

2.2. Synthesis of boron-containing cross-Linker (B-HEMA)

The boron-containing cross-linker (B-HEMA) was synthesized according to previously published literature with little modification [37]. A mixture of TMB (0.52 g, 5 mmol) and HEMA (2.05 g, 15.75 mmol) were dissolved in 60 mL of anhydrous acetonitrile, and the solution was stirred at $50\text{ }^\circ\text{C}$ over 3 h N_2 atmosphere. Subsequently, the reaction temperature was raised to $70\text{ }^\circ\text{C}$ and continued to stir over another 3 h to eliminate methanol that generated as byproduct. Finally, the resulting mixture was distilled at $60\text{ }^\circ\text{C}$ under reduced pressure to remove unreacted TMB and

residual acetonitrile, and further dried under high vacuum. B-HEMA was stored in an Argon filled glove box to prevent possible hydrolysis. The product was characterized by ^1H -NMR and ^{11}B -NMR spectroscopy. ^1H -NMR [400 MHz, CDCl_3 , δ (ppm), TMS ref] of B-HEMA: 6.17 (vinyl, CH), 5.62 (vinyl, CH), 4.31 ($\text{CH}_2\text{-O-C}$ (O)), 3.91 ($\text{CH}_2\text{-O-B}$), 1.98 (isobutyl, CH_3). ^{11}B -NMR [400 MHz, CDCl_3 , δ (ppm), TMS ref] of B-HEMA: 18.22 (B-O).

2.3. Preparation of plastic crystal polymer electrolyte film (PCPE)

Firstly, 1 M NaClO_4 was added into melted SCN at 60 °C under magnetic stirring to form a homogeneous solution. Subsequently, stoichiometric as-prepared B-HEMA and photo-initiator HMPP were added to the solution, wherein the concentration of B-HEMA was 20 wt% of the solution and the concentration of HMPP was 1 wt% of B-HEMA monomer. The PCN was chosen as mechanical skeleton and immersed to above mixture solution for 5 min, and then sandwiched between two clean glass plates. The final plastic crystal polymer composite film was obtained by exposing the glass plates to a Hg-UV lamp for 2 mins (400 W, 365 nm, the distance between lamp and glass plate upper-surface is 15 cm), and was denoted as B-PCPE. In addition, S-PCPE and C-PCPE films were fabricated as control experiments in this work, which were prepared using the same materials and procedure as used for the preparation of B-PCPE. The S-PCPE film was prepared without PCN mechanical skeleton and the C-PCPE was prepared using ETPTA as cross-linker instead of B-HEMA. All preparation process was performed in an argon-filled glove box.

2.4. Method

2.4.1. Sample characterization

Nuclear magnetic resonance (NMR) measurements were conducted on a Varian Mercury Plus 400MHz spectrometer with CDCl_3 as solvent and tetramethylsilane (TMS) as a reference at room temperature to confirm the structure of samples. Fourier transform infrared spectroscopy (FT-IR) analysis was performed on a spectrophotometer (Spectrum 100, Perkin Elmer, Inc., USA) a spectral resolution of 4 cm^{-1} . The morphology of the PCPE films was characterized by field emission

scanning electron microscopy (FESEM, Nova Nano SEM, FEI-450, USA). The energy dispersive X-ray spectroscopy (EDS) analysis technique was performed along with FESEM to confirm the presence of elements in the as-prepared samples. Differential scanning calorimeter (DSC) was carried out on Perkin-Elmer Pyris 1 USA from -60 to 120 °C at a heating/cooling rate of 10 °C/min under nitrogen atmosphere to evaluate the thermal behavior of the PCPEs. The stress-strain tests of the samples were measured using Instron 4465 instrument with a tensile speed of 5 mm/min.

2.4.2. Electrochemical evaluation

The ionic conductivity (σ , S/cm) of plastic crystal polymer electrolytes was measured by AC impedance spectroscopy using a Zennium Electrochemical workstation (Zahner Ennium, IM6ex) at varied temperatures ranging from 30 to 70 °C. The PCPE films (S-PCPE, C-PCPE, and B-PCPE) were sandwiched between two stainless-steel (SS) blocking electrodes, and the spectra were obtained in the frequency range from 0.1 Hz to 1 MHz with an AC amplitude of 10 mV. The ionic conductivity of the PCPE films is calculated based on Equation (1):

$$\sigma = \frac{d}{R_b \cdot S} \quad (1)$$

where R_b (Ω) is the bulk resistance of the samples obtained from the intercept at the real axis in the Nyquist plot, d (cm) is the thickness of the electrolyte films, and S (cm^2) is the sample area of the electrolyte/electrode interface. Linear Scanning Voltammetry (LSV) tests were tested on Na/PCPE/SS cells on the electrochemical workstation (Autolab, AUT30FRA2 ECICHEMIE) between 1.0 V and 6.5 V with a scan rate of 2 mV/s, where SS is a working electrode and Na metal is counter/reference electrode. Sodium-ion transference number (t_{Na^+}) was determined by the Bruce-Vincent-Evans method [38]. The PCPEs were sandwiched between two non-blocking sodium metal disks to form a symmetrical Na/PCPE/Na in 2032 coin cell, and the cell was polarized by a constant DC voltage of 10 mV. The t_{Na^+} from this method is calculated based on Equation (2):

$$t_{Na^+} = \frac{I_s R_f (\Delta V - I_0 R_f^0)}{I_0 R_i (\Delta V - I_s R_f^s)} \quad (2)$$

where ΔV is the DC voltage applied to the cell, I_0 and I_s are the initial and steady-state currents, R_i and R_f are the initial and final resistances of the electrolytes, and R_i^0 and R_f^S are the initial and steady-state interfacial resistance, respectively.

2.4.3. Cell Design and performance evaluation of all-solid-state SIBs

The $\text{NaNi}_{1/3}\text{Fe}_{1/3}\text{Mn}_{1/3}\text{O}_2$ (NFM) cathode for SIBs was prepared by mixing NFM active material, PVdF binder, and Super P at a weight ratio of 80:10:10. The mixture was dispersed in anhydrous NMP and stirred in a homogenizer to form homogeneous slurry. The slurry was casted onto the current collector of aluminum foil and was dried at 100 °C for 48 h under vacuum. The hard carbon (HC) anode for SIBs was prepared by mixing hard carbon (HC, a gift from Sinopoly Battery Research Center), PVdF binder, and Super P at a weight ratio of 80:10:10. Other preparation condition is the same as above NFM cathode. NFM and HC loading levels in both electrodes were 1.5-2.0 mg/cm². Subsequently, the final c-NFM cathode and c-HC anode were obtained by immersing the prepared NFM or HC electrodes in the B-HEMA (or ETPTA)/SN/NaClO₄ mixture at 60 °C overnight and then polymerized under ultraviolet irradiation with HMPP as photoinitiator. All-solid-state sodium-ion batteries (c-NFM/PCPEs/c-HC) were assembled in 2032 coin cells in argon-filled glove box with above c-NFM as cathode, c-HC as anode, and PCPEs (S-PCPE, C-PCPE, and B-PCPE) as solid electrolytes as well as separators. Charge/discharge tests of the c-NFM/PCPE/c-HC batteries were conducted at voltages of 1.5-3.8 V at room temperature (23 °C) with a C-rate of 1.0 C, where 1.0 C rate corresponds to a current density of 100 mA/g.

3. Results and discussion

3.1. Synthesis and Characterization of B-HEMA and PCPEs (S-PCPE, C-PCPE and B-PCPE)

B-HEMA was synthesized by the substitution reaction of trimethyl borate (TMB) and 2-hydroxyethyl methacrylate (HEMA). NMR was conducted to confirm the structure of B-HEMA and was shown in Fig. S1 and S2. Noticeably, compared with the ¹H-NMR spectrum of HEMA in Fig. S1, the resonance peak of d shifted

downfield when the hydroxyl groups of HEMA are substituted by the borate (from 3.87 ppm to 3.99 ppm), which is due to the electron deficient nature of the borate [39, 40]. This phenomenon confirms the successful incorporation of the borate into the HEMA. Furthermore, as shown in Fig. S2, ^{11}B -NMR was conducted to further analyze the structure of B-HEMA. Both TMB and B-HEMA exhibited resonance peaks at around 20.2 ppm which corresponded to the characteristic signal of the tri-coordinated boron atoms [39]. And compared with the spectrum of TMB, B-HEMA showed a much broader resonance peak due to the attached longer HEMA chains. All the noteworthy resonance peaks revealed in both ^1H -NMR and ^{11}B -NMR verify the successful synthesis of B-HEMA in this study.

As illustrated in Fig. 1A, B-PCPE was prepared by *in-situ* UV-curing of PCE and B-HEMA inside a porous PCN nonwoven skeleton. Meanwhile, one PCPE film prepared without PCN mechanical skeleton (donated as S-PCPE) and another prepared using trimethylolpropane ethoxylate triacrylate (ETPTA) as cross-linker instead of B-HEMA (C-PCPE) were obtained as control experiments through the similar method. It's important to note that the PCE used here is 1M NaClO_4 in SN, because the ionic conductivity of PCEs reached a highest value when the NaClO_4 concentration in SN is 1M, as shown in Fig. S3. Interestingly enough, with the increase in NaClO_4 concentration, the appearance state of PCEs changes from all-solid state to semi-solid state, and to entirely liquid state at last, as shown in Fig. S4. Although 1M NaClO_4 in SN used here shows liquid behavior, it will be completely solidified through the UV-crosslinking reaction of B-PCPE or ETPTA. The UV-crosslinking reaction of the PCPEs was examined by the FT-IR analysis. From Fig. 1B, the peak at 1635 cm^{-1} for PCE/B-HEMA mixture (S-PCPE before UV curing) is assigned to the characteristic vinyl bonds ($\text{C}=\text{C}$) of the B-HEMA cross-linker. It was found that the stretching vibration absorption of $\text{C}=\text{C}$ clearly disappeared in the spectra of S-PCPE, C-PCPE and B-PCPE after UV curing polymerization reaction, indicating that the methacrylate groups of ETPTA or B-HEMA cross-linker is successfully polymerized in the PCE and PCN nonwoven [41, 42]. In addition, the digital photographs of S-PCPE before and after UV curing

were also provided in Fig. S5. It can be seen that PCE/B-HEMA mixture before UV curing showed a liquid state at room temperature which can't maintain two-dimensional stability. After UV curing, the PCE/B-HEMA mixture became white solid state, namely S-PCPE. This further revealed the successfully polymerization of B-HEMA in PCE and indicated that the UV-crosslinked B-HEMA polymer can provide mechanical support to PCE.

3.2. Morphology and physical properties of PCPEs (S-PCPE, C-PCPE and B-PCPE)

The morphologies of PCN nonwoven and B-PCPE were investigated by SEM. It can be seen from Fig. 1C, PCN nonwoven was composed of randomly arranged nanofibers and it possessed numerous pores with larger pore size in the range of tens of micrometers, which would provide a favorable framework for supporting mechanically plastic crystal polymer electrolyte. Fig. 1D shows the SEM image of B-PCPE. It was noticed that the B-PCPE exhibited a homogenous and smooth surface, and the pores of the fibrous matrix of PCN nonwoven were filled with PCPE, revealing the successfully incorporation of PCPE into PCN nonwoven skeleton, which was of great significance to prevent possible micro-short circuiting. Meanwhile, compared to the pristine PCN nonwoven (the insert picture in Fig. 1C), the B-PCPE film (the insert picture in Fig. 1D) showed a translucent state, indicated that the PCPE is nearly compatible with the polymer chains in the PCN nonwoven and the bundles of polymer chains are effectively segregated by the PCPE. In order to examine the distribution of PCE and UV-curing B-HEMA inside the PCN nonwoven, energy dispersive X-ray mapping analysis was performed. Fig. 1E displays the cross-section SEM image of B-PCPE and the corresponding elements (C, O, Na, N, and B) mapping pictures. The uniform distribution of N, Na and B over the whole area of the material further suggests the presence and well-impregnation of PCE and UV-curing B-HEMA in PCN nonwoven. Film thickness of solid electrolytes affects the batteries' electrochemical performance to some extent. Generally, reduced film thickness can facilitate facile ionic transport, allow it impart higher ionic conductance and also contribute to higher volumetric energy density [43, 44]. It can be seen from Fig. 1E,

the thickness of B-PCPE was only about 100 μm , which was much smaller than that of pure S-PCPE without PCN nonwoven support (around 180-200 μm , Fig. S6).

Differential scanning calorimeter was conducted to investigate the plastic crystal behavior of B-PCPE. As shown in Fig. 1F, for SCN, there are two distinctive endothermic peaks were detected at $-35\text{ }^{\circ}\text{C}$ and $58\text{ }^{\circ}\text{C}$, corresponding to the crystal-to-plastic crystal transition temperature (T_{cp}) and the melting point (T_{m}) of SCN, respectively [45, 46]. When 1M NaClO_4 was added, an obvious decline in T_{m} and a slight change in T_{cp} were observed in PCE curve, which was similar to the situation reported in previous papers [20, 22]. Meanwhile, the S-PCPE, C-PCPE, and B-PCPE also showed similar thermal behavior to those of PCE, demonstrating that the introduction of UV curried B-HEMA polymer and PCN nonwoven support did not lead to the disruption of plastic crystal behavior of PCE. More specifically, a large decrease in T_{m} of SCN in all samples was detected in comparison with the pure SCN, reflecting the deterioration of crystalline and also expedition of isomer transition from gauche to trans-phase [10, 47, 48]. The high density of trans-phase in SCN isomer reveals high concentration of structural defects, which is of great significance to facilitate the ionic mobility, thereby contributes to high ionic conductivity [22]. In addition, different from SCN, several weak DSC bands in the temperature range of $-25\text{ }^{\circ}\text{C}$ to $5\text{ }^{\circ}\text{C}$, resulting from the solid-solid phase transitions in the plastic crystalline state [20].

As is known to all, outstanding mechanical properties of solid electrolytes played a critical role in preventing internal short circuits and avoiding self-discharge of a battery [49, 50]. The poor mechanical property of PCEs has always been an urgent issue which limits their further application, especially in large-scale energy storage system. Pure PCE (1 M NaClO_4 in SCN) is displayed a liquid characteristic (Fig. S4) under natural condition, and it can't be solely used as a freestanding electrolyte in rechargeable batteries. Digital camera images of B-PCPE were shown in Fig. 2A. It was noticed that both S-PCPE and B-PCPE were totally solidified and displayed a self-standing behavior compared to pure PCE. Additionally, the B-PCPE film showed superior flexibility and deformability and could be flexibly bended, even twisted

around a glass rod. However, the S-PCPE is excessively soft and susceptible to deform or fracture under low stress (Fig. S7). The tensile test was conducted for pristine PCN, S-PCPE, C-PCPE, and B-PCPE to further evaluate their mechanical properties. As shown in Fig. 2B, the maximum stress of S-PCPE is only 1.71 MPa, which was harmful to safety and reliability of solid batteries. After the introduction of PCN support, the maximum stress of C-PCPE and B-PCPE were greatly improved compared with S-PCPE and reached 31.1 and 28.5 MPa, respectively, confirming that the original advantages of the PCN support (38.1 MPa, the slight decrease for C-PCPE and B-PCPE should be ascribed to high plasticity of SCN). Notably, the tensile stress of B-PCPE is much superior to reported works in rechargeable batteries as shown in Table S1. These superior mechanical properties could lower the possibility of the mechanical rupture and enhance the safety characteristics of the sodium-ion batteries. In view of the practical application, C-PCPE and B-PCPE could provide more reliable mechanical property and subsequently relieve the possibility of short circuit of the batteries, thus contribute to the safety and reliability of sodium-ion batteries [51].

3.3. *Ionic conductivity, electrochemical stability, and ionic transference number*

Fig. 2C illustrated temperature dependency comparison of the ionic conductivity of B-PCPE and C-PCPE, and S-PCPE at various temperatures. It was observed that all samples exhibited superior room-temperature ionic conductivity of more than 10^{-4} S cm⁻¹ owing to high concentration of PCE. Obviously, both B-PCPE and C-PCPE show the lower ionic conductivity than the S-PCPE over a wide range of temperature. This is mainly ascribed to the presence of the PCN nonwoven skeleton who has no ionic conducting ability. Furthermore, it should be noticed that C-PCPE containing no boron moieties provided a little higher ionic conductivity than B-PCPE. This should be explained by the Lewis acidity of the elements of B in B-PCPE. In fact, most electrolytes are dual-ion conductor in which both cation and anion are mobile, therefore, the ionic conductivity obtained is the total conductivity including cation (Na⁺) and counter anion (ClO₄⁻). As illustrated in Scheme 1, in B-PCPE, boron acted as an anion acceptor whose role is similar to that of single ion conductors, where

anions are partly immobilized and cations became the main or even single effective mobile ions, resulting in decrease in ionic conductivity [25, 52, 53]. However, C-PCPE is a typical dual-ion conductor where cation and anion migrate simultaneously, thus a higher ionic conductivity was obtained. Nevertheless, the B-PCPE still provided a satisfactory ionic conductivity of 0.36 mS/cm at room temperature, which is higher than that of polymer-based solid electrolytes of previous publications (As shown in Table S2).

The electrochemical stability window of the PCE and PCPEs was examined by linear-sweep voltammetry (LSV) with stainless steel as the working electrodes and sodium metal as the counter and reference electrodes. It can be seen from Fig. 2D that both PCE and S-PCPE show poor electrochemical stability, and obvious current peaks occurred after 3.4 V (vs Na/Na⁺), which is consistent with the result reported in previous publication [20]. However, there seems to be insignificant current peaks were detected in the C-PCPE and B-PCPE below 5.0 V (vs Na/Na⁺), thus providing an electrochemical window wide enough to allow the safe application of the electrolytes in SIBs with the selection of a great number of high-voltage cathode materials [54-56].

Apart from ionic conductivity and electrochemical stability, the sodium ion transference number (t_{Na}^+) is also one of the indispensable parameters for evaluating the performance of electrolytes. It's well known that the introduction of boron moieties can enhance the lithium ion transference number (t_{Li}^+) [28, 57, 58], but there is no report on t_{Na}^+ yet. Here, the effect of boron moieties on t_{Na}^+ in PCPEs was confirmed using the method proposed by Vincent and Evans [38], and the results of electrochemical impedance spectroscopy (EIS) and DC polarization tests (at room temperature) are given in Fig. S8 and Table S3. It was found that the value of t_{Na}^+ for C-PCPE is 0.26, which is much lower than that of B-PCPE, revealing dominant mobility of anions in C-PCPE, which is similar to the result in conventional liquid electrolytes. More importantly, compared to C-PCPE, B-PCPE deliver the higher t_{Na}^+ values of 0.62 as expected, indicative of partly single sodium-ion conducting behavior. But it's worth noticing that the value of t_{Na}^+ for B-PCPE may be further improved by

changing the B-HEMA loading in the B-PCPEs. The distinctively improved t_{Na}^+ values for B-PCPE should be attributed to the anion-capturing ability of boron moieties where boron atom with sp^2 hybridization possesses an unoccupied orbital, acting as Lewis acid to accept electronegative ClO_4^- group, as illustrated in Scheme 1A, Namely, the movement of ClO_4^- in the B-PCPE was restrained greatly, which have been demonstrated by Xiaohua Ma et al. in previous work [36]. In addition, for most solid electrolyte system, high cation transference number generally leads to quite poor room-temperature ionic conductivity, and usually in the range of 10^{-8} - 10^{-7} S/cm [59], which greatly limit their practical application. However, the B-PCPE synthesized in this work simultaneously possessed superior room-temperature ionic conductivity and high sodium ion transference number due to the presence of high-concentration PCE and anion-capturing boron moieties, which is beneficial for reducing the cell polarization and improving the cell performances.

3.4. *In situ growth of PCPE in electrodes*

In all-solid-state batteries, facile electrode-electrolyte interface contact and favorable ion transport between active material particles inside electrode play a significant role in cell performance. As illustrated in Fig. S9, in general all-solid-state battery, electrodes composed of active material, conductive agent and polymer binder or electrolytes, the surface area of the active materials is hard to fully utilize because the high viscosity of the polymer with high molecular weight hinders the successful penetration of ions into the small pores of the electrodes [60]. On the other hand, conventional all-solid-state cell was assembled in simple sandwich model where the electrolyte is kept as a film between the electrodes, severe interfacial impedance due to the harsh solid-solid contact between electrode and electrolyte existed, which will lead to poor ion transport in the electrode-electrolyte interface and further worsen cell performances. Therefore, the configuration optimization of electrodes as well as the construction of a “gentle” electrode/electrolyte interface with small interfacial resistance is of great importance to improve all-solid-state battery performance. In this study, c-NFM cathode and c-HC anode all containing PCPEs were designed by allowing in situ generation of plastic crystal polymer electrolyte inside the nanopores

of the electrodes to facilitate ion transport in electrodes and simultaneously form favorable electrode-electrolyte interface. Briefly, the cathode/anode for all-solid-state sodium-ion batteries were prepared by mixing NFM (prepared according to our previous work [61])/Hard carbon, Super P conductive carbon, and polyvinylidene fluoride (PVdF) as a binder to form a slurry using N-methyl-2-pyrrolidone (NMP) and then coated on aluminum current collectors with the mass loading of active materials of 1.5-2.0 mg/cm². Subsequently, the as-prepared NFM cathode or HC anode were immersed in the B-HEMA (or ETPTA)/SCN/NaClO₄ mixture at 60 °C overnight and then polymerized under UV lamp to yield the resulting c-NFM cathode or c-HC anode. And the preparation scheme is shown in **Fig. 3A**.

The X-ray diffraction patterns of the NFM powder, NFM and c-NFM cathodes are shown in Fig. S10. It can be seen clearly that all the peaks can be indexed to the NFM [61], implying that the PCPE maintains very good chemical stability against NFM active material. Both the conventional NFM cathode (Fig. 3B) and HC anode (Fig. 3D) had rough surfaces with micro-porous structures, in which a lot of pores with various sizes (0.5 to 10.5 μm for NFM cathode and 1 to 8.2 μm for HC anode) were presented between the active material particles. However, as seen in Fig. 3C, E, after *in situ* growth of PCPE in electrodes, the c-NFM cathode and c-HC anode with the PCPEs had a relatively smooth surface and the pores inside the composite electrodes were successfully filled with PCPEs, implying that the PCPEs were well impregnated and that the active particles were completely covered by the PCPEs. Furthermore, the EDS mapping of boron atoms and nitrogen atoms for c-NFM and c-HC electrodes (Fig. 3C and E) from PCPEs also demonstrated a uniform distribution of PCPEs in the electrode surfaces as well as complete impregnation into the interspaces. In addition, SEM images of cross-sectional of electrodes in Fig. S11 further verified the complete impregnation of PCPEs into active particles. The schematic of the electrodes with and without PCPE are shown in Fig. S12. In a conventional NFM cathode, because of the existence of numerous nanopores and interspaces derived from high viscosity of the polymer electrolyte, Na⁺ transfer generally occurs at a point contact interface where cathode particles are densely

connected together, and Na^+ transfer in the interspace (red circle regions in Fig. S11A) is nearly blocked. Thus, the efficiency of Na^+ transfer in a conventional NFM cathode is very low. The *in situ* generation of PCPE in c-NFM cathode can enhance the ion transfer efficiency. In c-NFM cathode, Na^+ transfer can occur at the point contact between cathode particles as well as the interspaces in which the PCPE with high ion conducting ability is filled (Fig. S11B), presenting a similar behavior to that of liquid electrolyte based cathodes. It is guessed that the PCPE *in situ* generated c-NFM cathode and c-HC anode will substantially contribute to cell performances.

3.5. Application in all-solid-state sodium-ion batteries at room temperature

All-solid-state sodium-ion batteries using c-NFM as cathode, c-HC as anode and PCPEs as electrolyte were assembled and the electrochemical performance of batteries were evaluated at room temperature (23 °C). For comparative studies, all-solid-state c-NFM / c-HC cells using S-PCPE and C-PCPE as electrolytes were also assembled. Fig. 4 A, B and C present the charge/discharge profiles of c-NFM/c-HC cells assembled with B-PCPE, C-PCPE and S-PCPE, respectively at 0.1 C in the potential range of 1.5-3.8 V. The battery assembled with B-PCPE shows a high first discharge capacity of 104.8 mAh g⁻¹ with Coulombic efficiency of 94.1%. Obviously, the B-PCPE exhibits the superior charge/discharge capacity retention capacity with cycling, as compared to the S-PCPE and C-PCPE showing gradual decrease in the charge/discharge capacities, and the cycle performances of these three kinds of electrolytes are summarized in Fig. 4D. It can be observed clearly that the B-PCPE provides high discharge capacity retention of about 80.1% after 120 cycles. For S-PCPE and C-PCPE, although they deliver acceptable first discharge capacities of 108.2 mAh g⁻¹ and 101.2 mAh g⁻¹, respectively, the discharge capacities distinctly decreased after five cycles, especially for S-PCPE, the discharge capacity began to sharply decline after 20 cycles. The higher first discharge capacity should be ascribed to the higher ionic conductivity of S-PCPE (Fig. 2C). It should be noticed that thickness of electrolyte film is also a key factor to determine cell performance, which behaves even more noticeable in all-solid-state battery. The B-PCPE and C-PCPE with flexible PCN support possess lower film thickness, this shorten Na^+ diffusion

path, thus contribute to facile ion transport, leading to better discharge capacity retention and cycle performance. What's more, it is reasonably guessed that the B-PCPE/C-PCPE may have better interfacial contact stability with electrodes due to its superior mechanical flexibility (Fig. 2A, B) [62], which can more effectively resist the volume change of electrodes during charge/discharge process and enable a long-term interfacial stability to maintain, thereby alleviating interfacial resistance between the B-PCPE/C-PCPE and electrodes during cycling [23]. As can be seen from Fig. S13, the surface morphology of the c-NFM cathode didn't change after cycling 120 cycles at the current density of 0.1 C at room temperature, which further verify the interfacial contact stability of B-PCPE to electrode.

Besides the cycle stability, a good rate capability is also a critical factor for practical application of all-solid-state SIBs. The rate capability of the c-NFM/B-PCPE/c-HC cell was examined by increasing the C-rate from 0.1 to 3.0 C for every five cycles at room temperature. In order to reliably verify the feasibility of B-PCPE in improvement of rate capability, the rate capability of c-NFM/c-HC cell using C-PCPE was also tested. Fig. S14 presents the corresponding charge/discharge profiles of c-NFM/B-PCPE/c-HC and c-NFM/C-PCPE/c-HC cells at various current densities, respectively. The discharge capacities of cells gradually decrease with an increase of current density, indicating the electrode polarization loss of cells. Remarkably, although the discharge capacity value gradually decreases as the C-rate increases, B-PCPE shows relatively higher capacity values than C-PCPE over a wide range of current densities, and this difference in the discharge capacities becomes more noticeable at higher current densities, where contribution of ionic transport to the electrode polarization is more apparent [63, 64]. Fig. 4E summarizes the discharge capacities of the B-PCPE and C-PCPE as a function of current density, revealing the superior rate capability of the B-PCPE as compared to C-PCPE. It can be seen clearly that no obvious discharge capacity degradation was observed for c-NFM/B-PCPE/c-HC cell when the current density was less than 1 C, and the reversible discharge capacity still maintained 80.24 mAh g^{-1} at a high current density of 3 C. Moreover, the recoverable capacity is 95.1% when the current density turns

back to 0.1 C, indicating the excellent reversible capacity recovery behavior of B-PCPE. Whereas, by contrast, C-PCPE only displayed an acceptable rate capability at low current density, and the discharge capacity sharply decreased at higher current density (> 0.5 C) due to the aggravated electrode polarization, and the reversible discharge capacity of C-PCPE was less than 50 mAh g^{-1} at the current density of 3 C. This superior rate capability of B-PCPE can be reasonably explained by the higher Na^+ transference number of B-PCPE (0.62 vs. 0.27) derived from the anion-trapping boron moieties and relatively large ionic conductance of the B-PCPE. As illustrated in Scheme 1, C-PCPE is a typical dual-ion conducting system where ClO_4^- and Na^+ move towards opposite directions, this will cause concentration gradients of the salt and cell polarization during the charge/discharge process, resulting in deterioration of long term cycle performance and rate capability. While ClO_4^- was partly trapped by boron atom in B-PCPE, thus exhibits a similar single Na^+ conducting behavior during the charge/discharge process, this can significantly alleviate concentration polarization of electrolyte during the charge/discharge process, thus facilitates effective Na^+ transport, leading to the improvement of discharge capacity and also discharge rate capability [26, 65]. It is clearly seen that the B-PCPE containing anion-trapping boron moieties shows much better cycling performance and rate capability at room temperature as compared to the C-PCPE, indicating that B-PCPE has promising application in all-solid-state sodium-ion batteries.

4. Conclusion

In summary, a nonwoven supported plastic crystal polymer electrolyte containing anion-trapping boron moieties (B-PCPE) was successfully prepared by *in situ* growth technique for all-solid-state sodium ion battery applications. The B-PCPE shows excellent room temperature ionic conductivity (0.36 mS cm^{-1}) and greatly improved sodium transference number (0.62) due to the presence of anion-trapping boron moieties. Owing to the nonwoven based three dimensional cross-linked network structure, the B-PCPE exhibited superior mechanical property with a high tensile strength of 28.2 MPa. Furthermore, PCPE permeated composite NFM cathode

(c-NFM) and HC anode (c-HC) were prepared by *in situ* generation of PCPE in electrodes to form favorable electrode/electrolyte interfaces. The c-NFM/c-HC all-solid-state sodium ion battery using B-PCPE as electrolyte presented superior cycling stability and rate capability at room temperature, indicating great potential of this B-PCPE in practical application in all-solid-state SIBs. Most significantly, the *in situ* growth technique of PCPE in electrodes illustrated in this work may provide a universal platform to facilitate effective ion transport inside electrodes as well as electrode/electrolyte interfaces in other all-solid-state battery systems.

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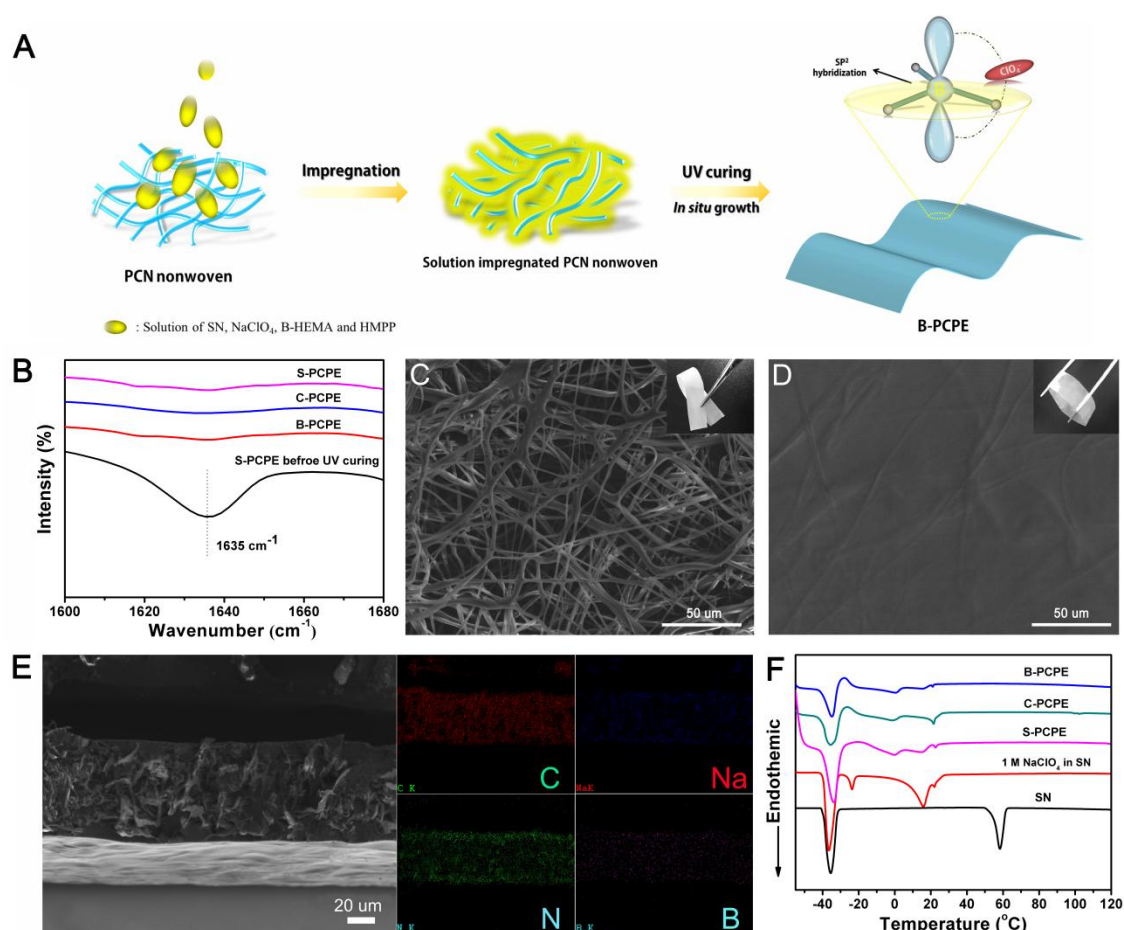


Fig. 1. A) Schematic illustration of the synthesis procedure of B-PCPE. B) FT-IR spectra of acrylic C=C double bonds of ETPTA in C-PCPE, or B-HEMA in B-PCPE and S-PCPE before/after UV polymerization. C) Typical SEM image of PCN nonwoven, the inset is the digital photograph of PCN nonwoven. D) Typical SEM image of B-PCPE, the inset is the digital photograph of B-PCPE. E) SEM image of the fracture of B-PCPE and corresponding EDAX elemental mapping (C, Na, N, and

B) on the fracture of B-PCPE. F) DSC profiles for SN, PCE, S-PCPE, C-PCPE, and B-PCPE.

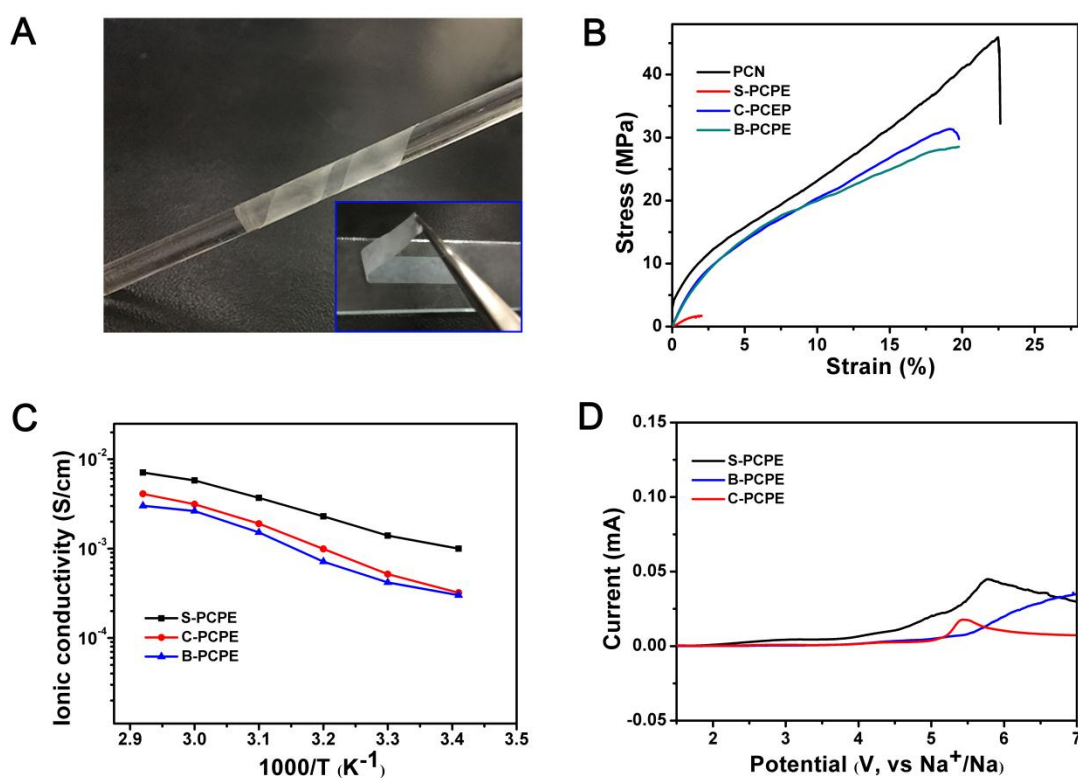
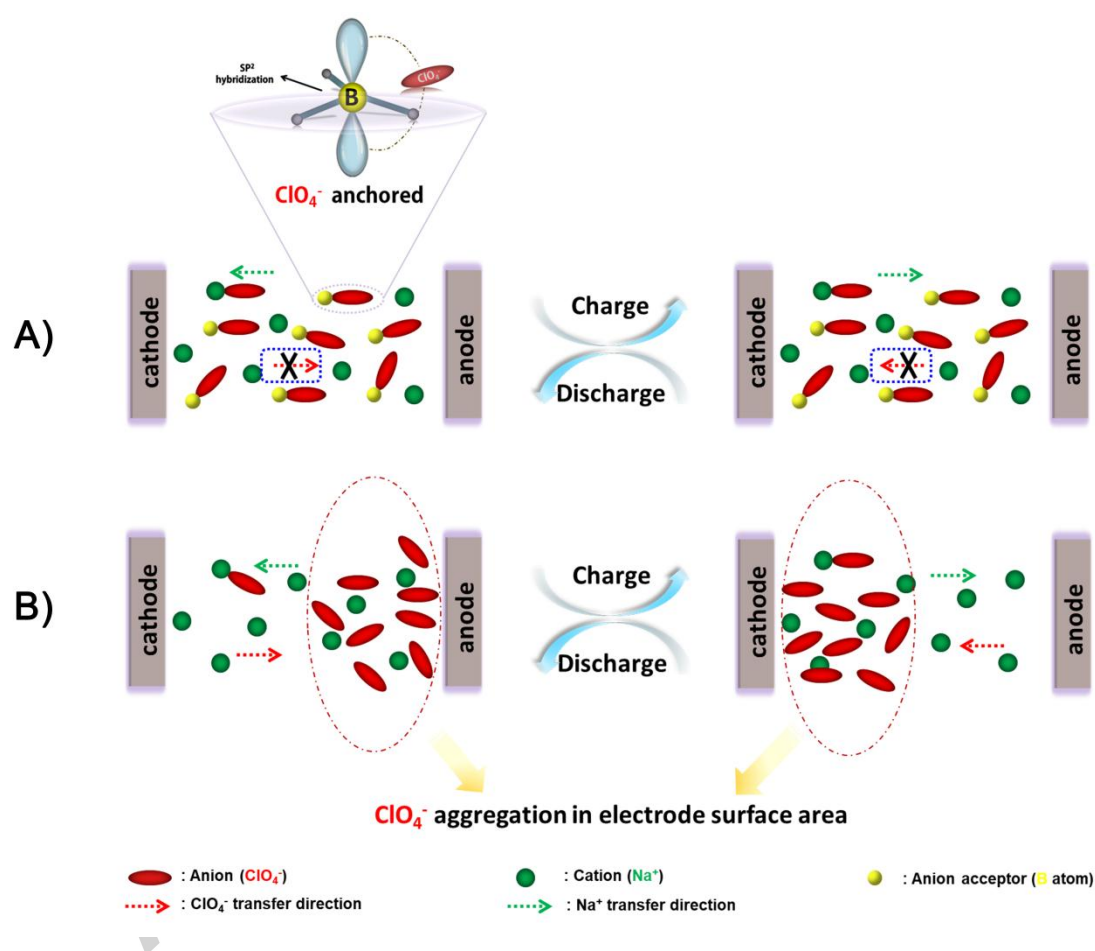


Fig. 2. A) The digital photograph of flexibility display for B-PCPE. B) Stress-strain curves for PCN, S-PCPE, C-PCPE, and B-PCPE. C) Temperature dependency comparison of ionic conductivity. D) LSV curves of S-PCPE, C-PCPE, and B-PCPE.



Scheme 1. A) Illustration of similar single-ion transport behavior in the B-PCPE and B) conventional dual-ion transport behavior in the C-PCPE during charge-discharge process.

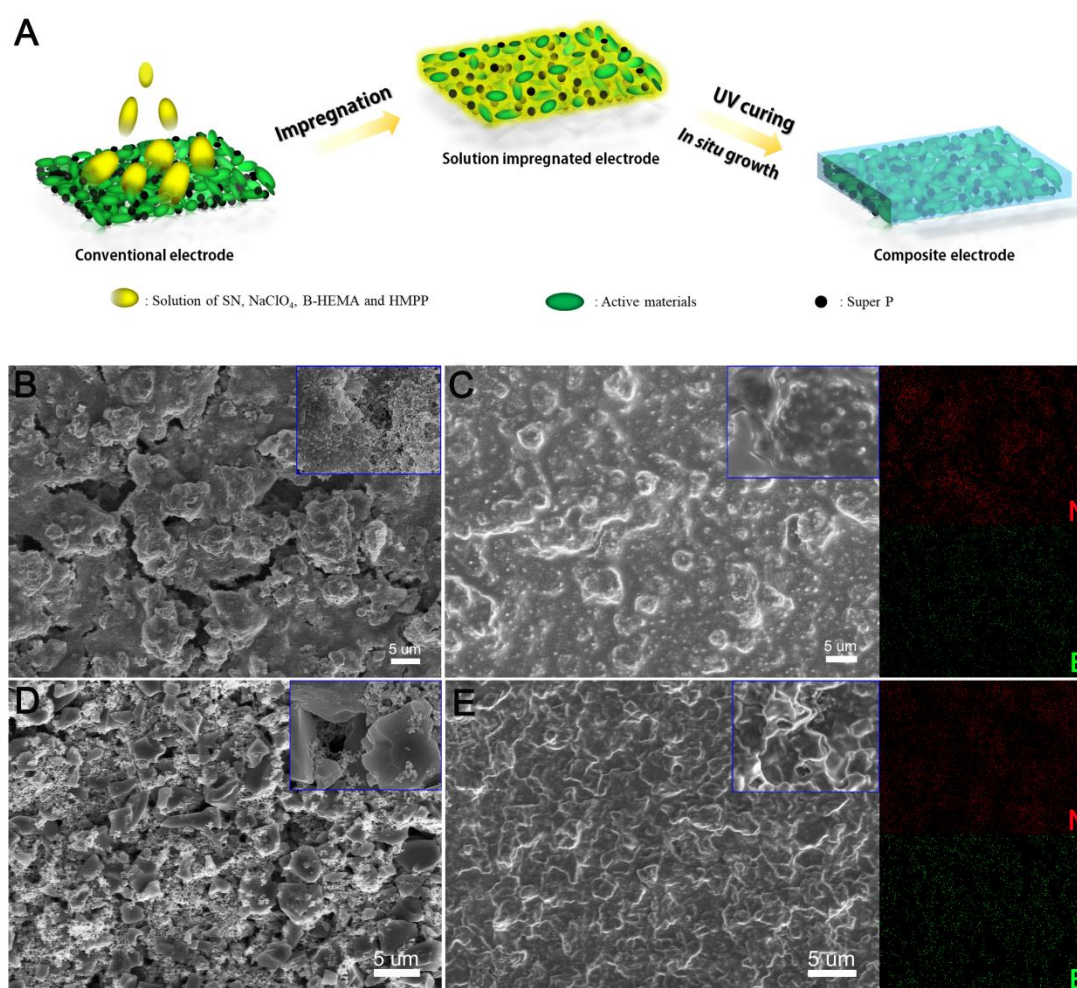


Fig. 3. A) Illustration of the *in situ* generated composite electrodes. B) SEM image of conventional NaNFM cathode. C) SEM image of c-NaNFM cathode and corresponding EDAX elemental mapping (N and B). D) SEM image of conventional HC anode. E) SEM image of c-HC anode and corresponding EDAX elemental mapping (N, and B).

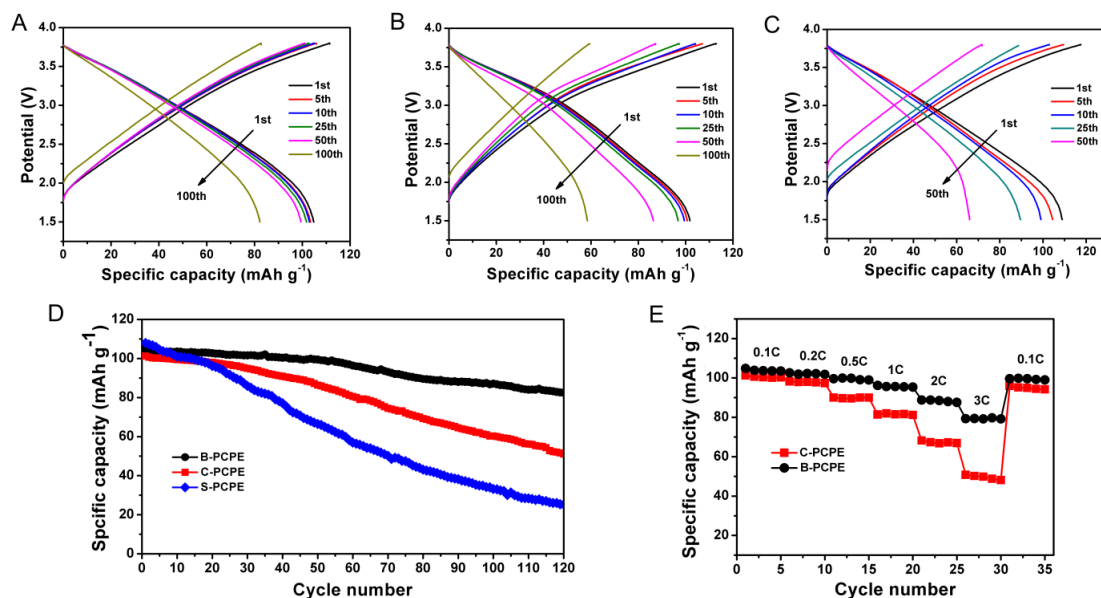


Fig. 4. Galvanostatic charge-discharge profiles of A) B-PCPE, B) C-PCPE and C) S-PCPE. D) Cycling performance comparison of B-PCPE, C-PCPE, and S-PCPE at 0.1 C at room temperature. D) Comparison of rate capability between B-PCPE and C-PCPE at various discharge current densities (from 0.1 C to 3 C) at room temperature.